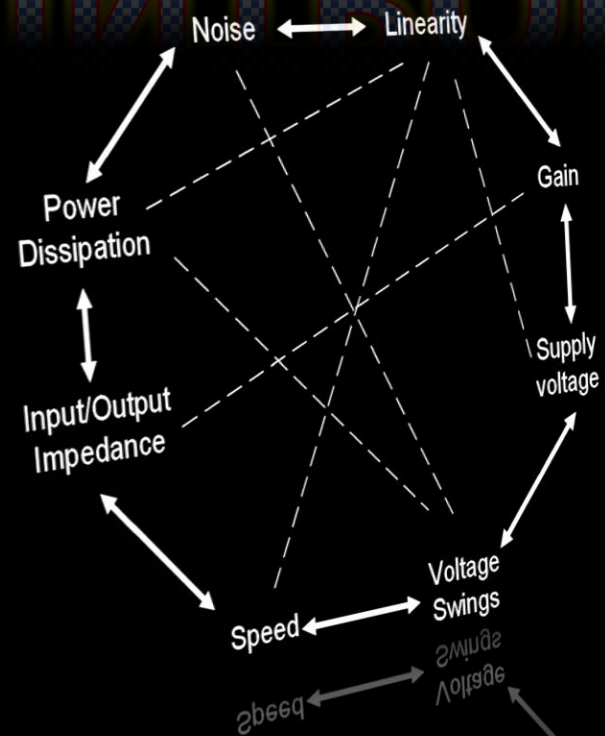


INTRODUCTION



TO

ANALOG

DESIGN



Er. Sailesh Mishra

CHIP DESIGN ENGINEER

(Sevya Multimedia Technologies)

B.Tech (ECE | Silicon University)

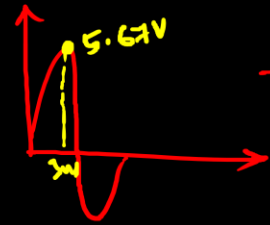
Specialization in Microelectronics & VLSI

$5.67 \rightarrow 6 > 5 ; 5.67 \rightarrow 6$
 $5.67 \rightarrow 6 > 5 \checkmark ; 5.7 \rightarrow 6$
 $5.67 \rightarrow 7 > 5 \checkmark$
 (OFF) $\rightarrow 0.C$
 FALSE $\leftarrow 0 \rightarrow$ Binary zero and $1 \rightarrow$ Binary one $\rightarrow$ TRUE
 (ON) $\rightarrow 5.C$

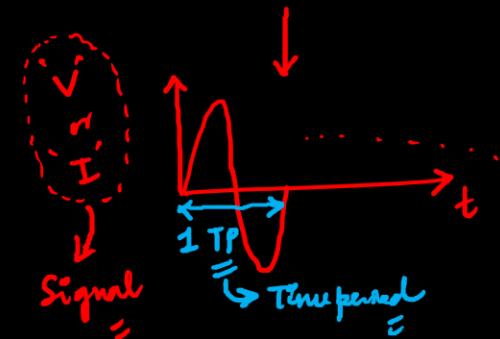
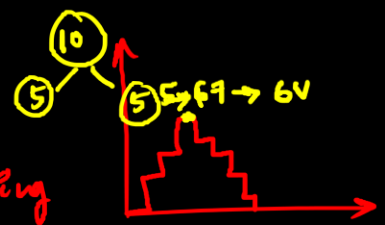
DIGITAL

- PUN PDN (1)
- (2)
- pseudo NAOG / PNOG (3)
- domino dynamic nMOS (4)

Continuous time domain system



ADC / Encoding - decoding + Quantizing



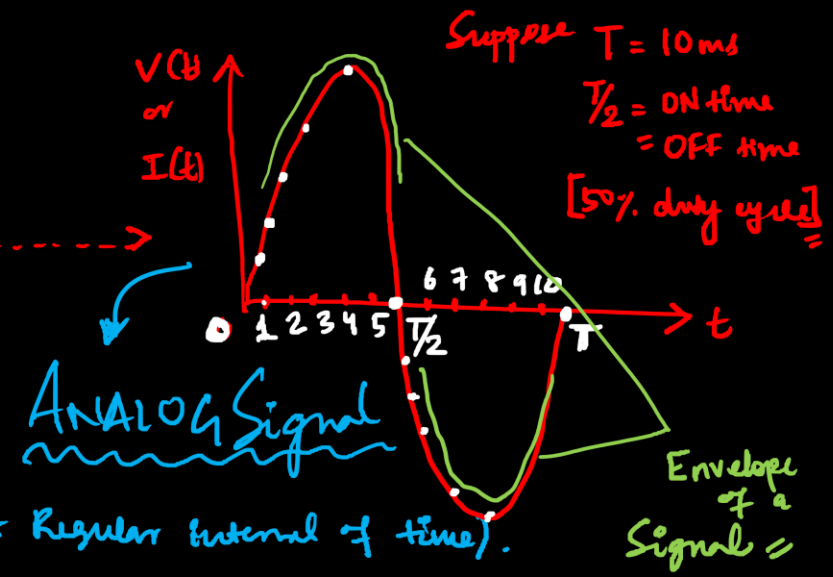
[Mathematical Function]

eg $\rightarrow 2\pi f_c \cdot t$

$\sin(2\pi f_c \cdot t), \cos(2\pi f_c \cdot t)$

HARMONIC FUNCTION $\rightarrow$ SHM $\rightarrow$ PERIODIC (repeats at Regular interval of time).

$f_c = \frac{1}{T} \rightarrow$ time period



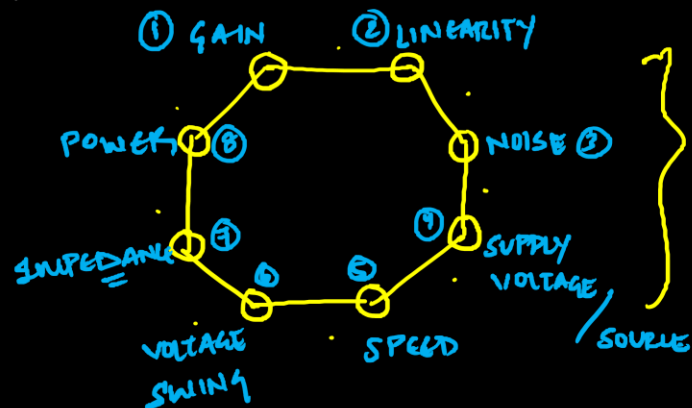
Suppose $T = 10ms$

$T/2 =$ ON time
 $=$ OFF time

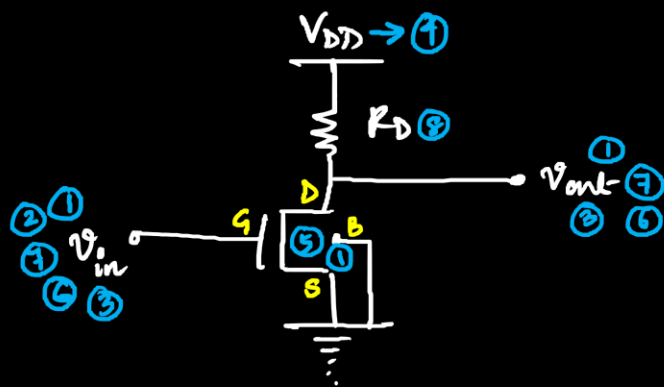
[50% duty cycle]

Envelope of a Signal

ANALOG DESIGN OCTAGON :-



These are the Salient features of any ANALOG DESIGN.



→ RESISTIVE LOAD ←
Common Source Stage Amplifier

→ To Implement & Execute a MOSFET as an Amplifier;
We Need to BIAS it / OPERATE it in "Saturation" !!

→ MOSFET { PMOS } → E_S^N → PMOS.

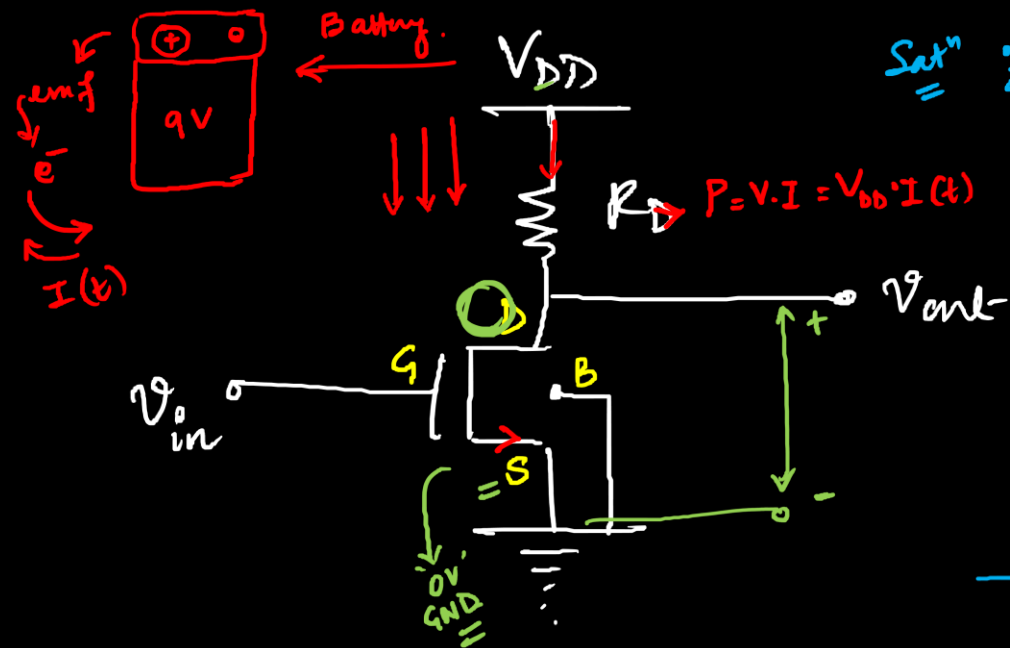
$$V_{SD} \geq V_{SG} - |V_{tp}|$$

NMOS.

→ WHY SATURATION? $V_{DS} > V_{GS} - V_{tn}$

Because "in Sat"; MOSFET have Better Current driving Capability.

⊙ MATHEMATICAL ANALYSIS (P.T.O)



$$Sat^n \div [V_{DS} > V_{GS} - V_{th} - I_D \cdot R_D]$$

or
Terminal
voltage
wrt
GND
= or
Source

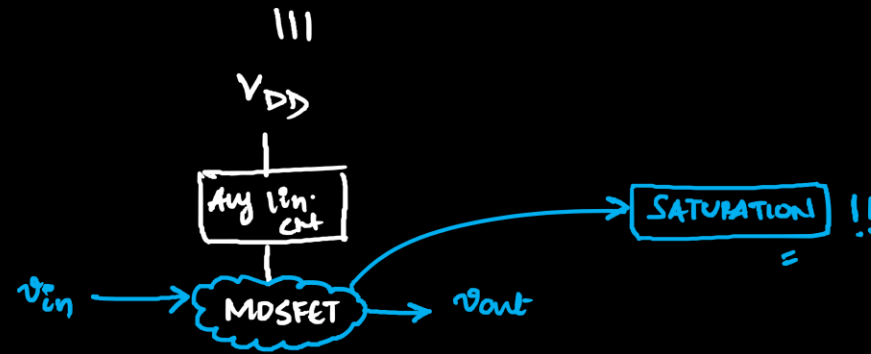
is
Basically
the
i/p
signal
or
voltage.
= i.e. (v_{in})

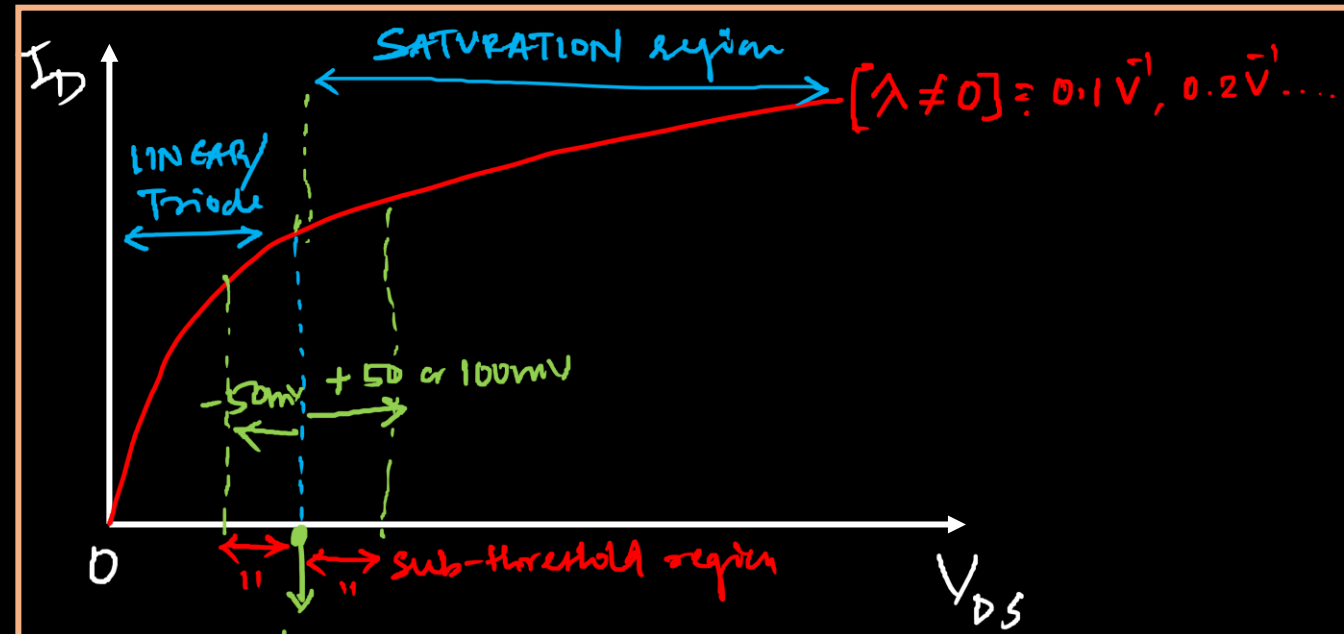
Threshold
voltage of
the device
always
[constant]

$$V_{DS} > V_{GS} - V_{th} - I_D \cdot R_D$$

"Equal to" physically means, I am
operating MOSFET @ edge of linear & Satⁿ.

low
power VLSI design !!





(MARGIN of linear region & Satⁿ region).

► Example 2.2

On a V_{DS} - V_{GS} plane, show the regions of operation of an NMOS transistor.

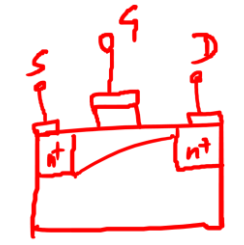
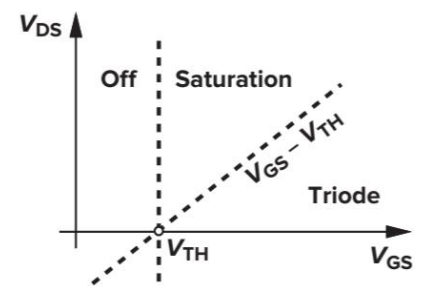
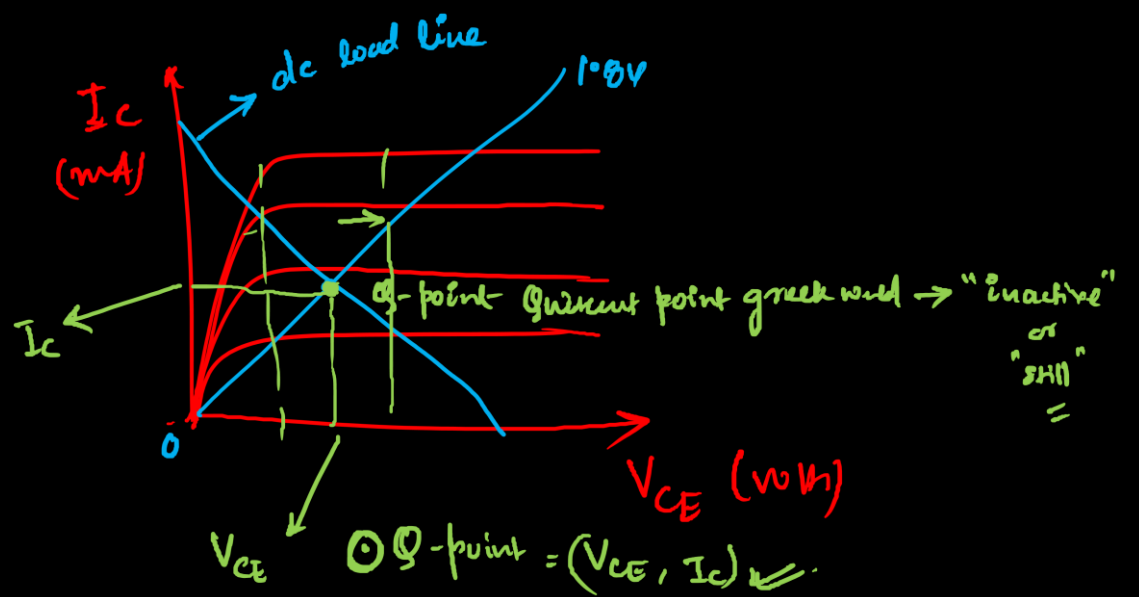


Figure 2.18 V_{DS} - V_{GS} plane showing regions of operation.

Solution

Since the value of V_{DS} with respect to $V_{GS} - V_{TH}$ determines the region of operation, we draw the line $V_{DS} = V_{GS} - V_{TH}$ in the plane, as shown in Fig. 2.18. If $V_{GS} > V_{TH}$, then the region above the line corresponds to saturation, and that below the line corresponds to the triode region. Note that for a given V_{DS} , the device eventually leaves saturation as V_{GS} increases. The minimum allowable V_{DS} for operation in saturation is also called $V_{D,sat}$. It is important to bear in mind that $V_{D,sat} = V_{GS} - V_{TH}$.



HOW TO OPERATE MOSFET IN SATURATION ?

$$\rightarrow V_{th} = 600\text{mV} = 0.6\text{V}$$

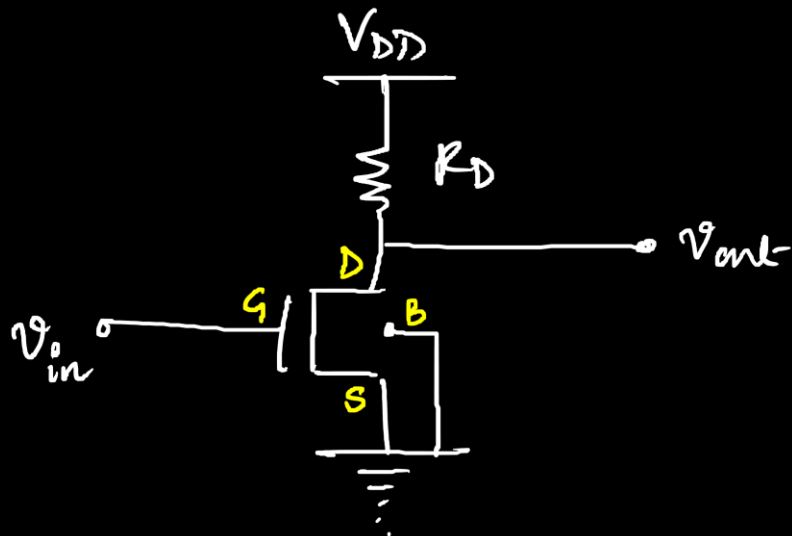
→ CONSTANT !!

$$\rightarrow V_{gs} = V_{th} + 200\text{mV} =$$

$$600\text{mV} + 200\text{mV} = 800\text{mV} = \boxed{0.8\text{V}}$$

INPUT / Calibration
GATE
voltage =

$$\rightarrow V_{ds} = V_{gs} + 210\text{mV} = 800\text{mV} + 210\text{mV} = 1010\text{mV} = \boxed{1.01\text{V}}$$



$$\text{SAT}^n \rightarrow V_{ds} > V_{gs} - V_{th}$$

$$= 1.01\text{V} > 0.8 - 0.6\text{V}$$

$$= 1.01\text{V} > 0.2\text{V} \checkmark \rightarrow \text{SATURATION.}$$

So; From the Above Analysis of Bias point Calibration;

V_{ds} is Much More; Hence; Power Consumption will be proportionally More !! ; So; We'll decrease it by Keeping a Margin of $\approx 50-100\text{mV}$ or 200mV .

$$(V_{ds})_{\text{New}} = (V_{gs} - V_{th}) + 200\text{mV} = 0.2 + 200\text{mV} = \boxed{0.4\text{V}} !!$$

Now; Finally we've calibrated the device with the following Terminal Potentials as Below:-

(i) GATE VOLTAGE $\rightarrow V_{in} \rightarrow V_{GS} = 0.8V$ ✓

(ii) DRAIN VOLTAGE $\rightarrow V_{out} \rightarrow V_{DS} = 0.4V$ ✓

$$\begin{aligned} (I_D)_{sat} &= \frac{1}{2} \mu_n C_{ox} \cdot \frac{W}{L} \{V_{GS} - V_{th}\}^2 \\ &= \frac{1}{2} \times \frac{200 \mu A}{V^2} \times \frac{10}{1} \{0.8V - 0.6V\}^2 \\ &= 0.5 \times \frac{200 \mu A}{V^2} \times 10 \times (0.2)^2 \end{aligned}$$

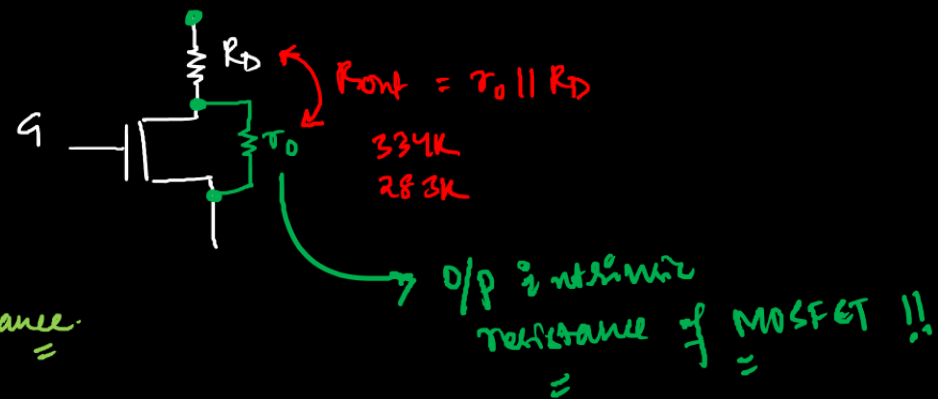
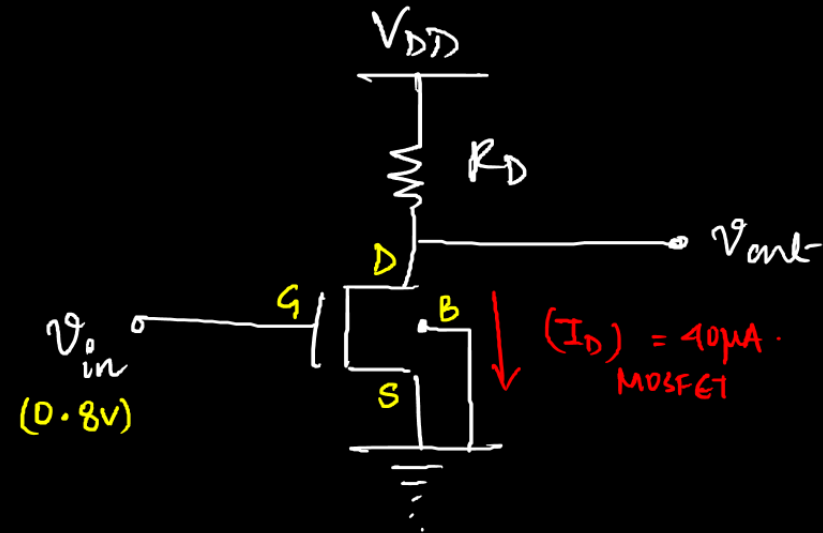
$$\Rightarrow \boxed{(I_D)_{sat} = 40 \mu A} =$$

We know;

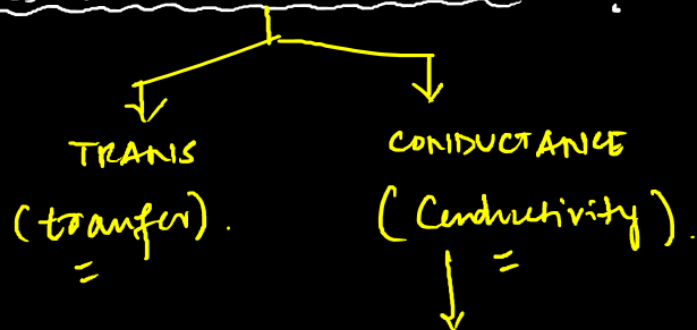
$$A_v = g_m \cdot R_D \text{ or } [g_m] R_{out}$$

$\rightarrow$ MOSFET's transconductance.

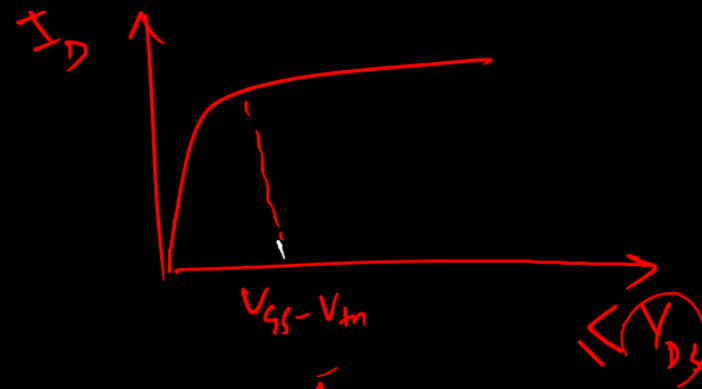
① $\underline{g_m} \rightarrow$ total ckt's TRANSCONDUCTANCE.



What is TRANS CONDUCTANCE ?



the extent/ability to conduct flawlessly.



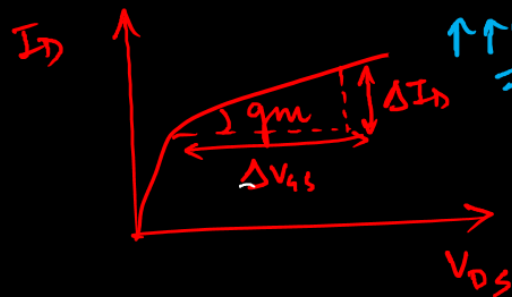
① (Current detecting capability)

$$\Rightarrow A_v = g_m \cdot R_{out}$$

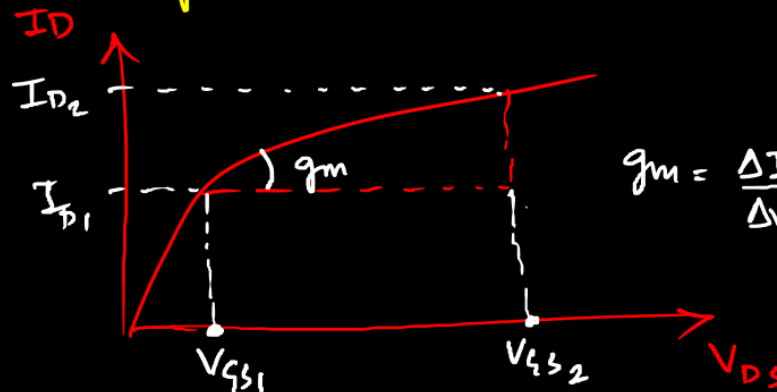
② ① $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$

d/p voltage

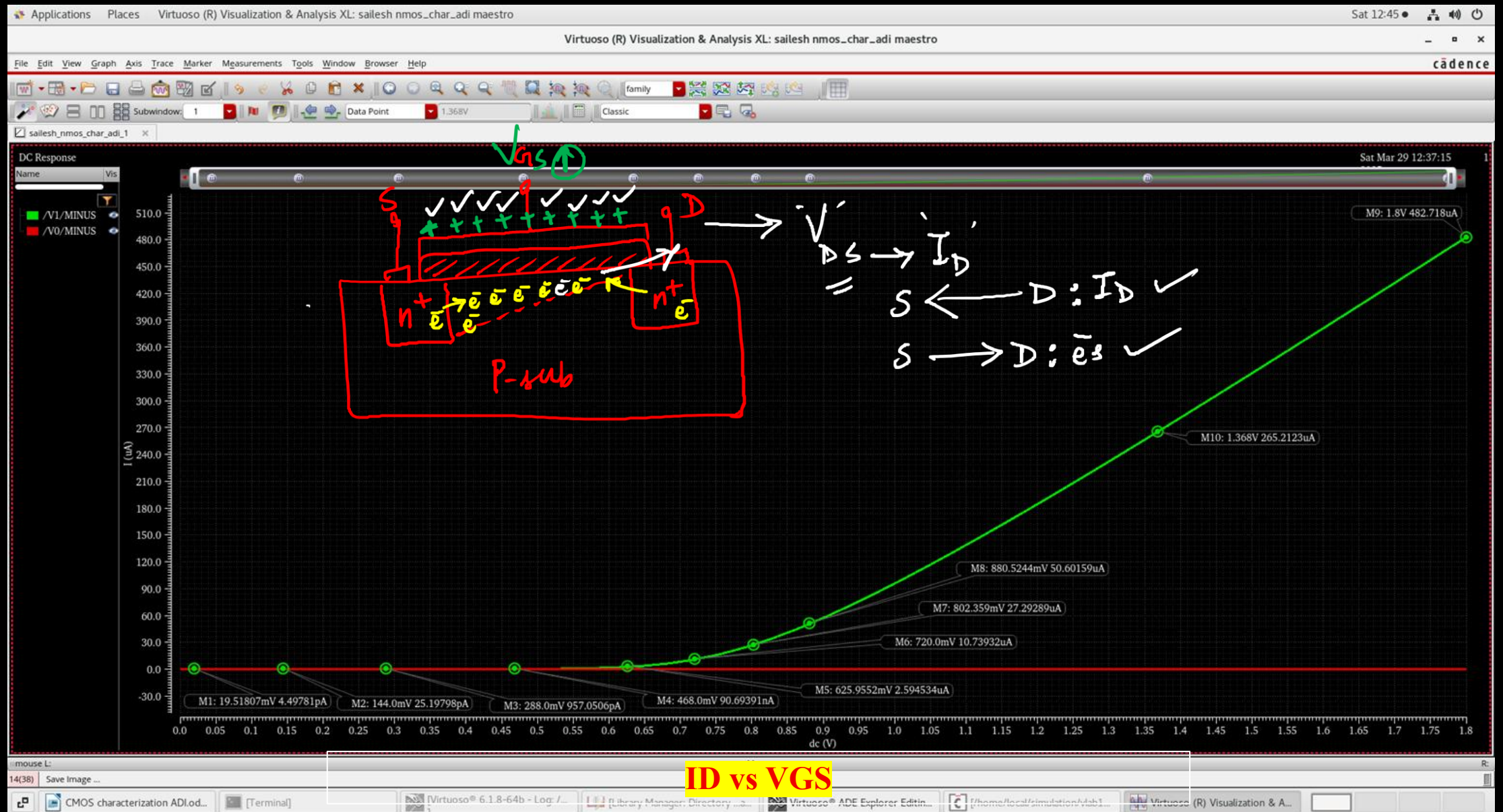
↑↑↑↑

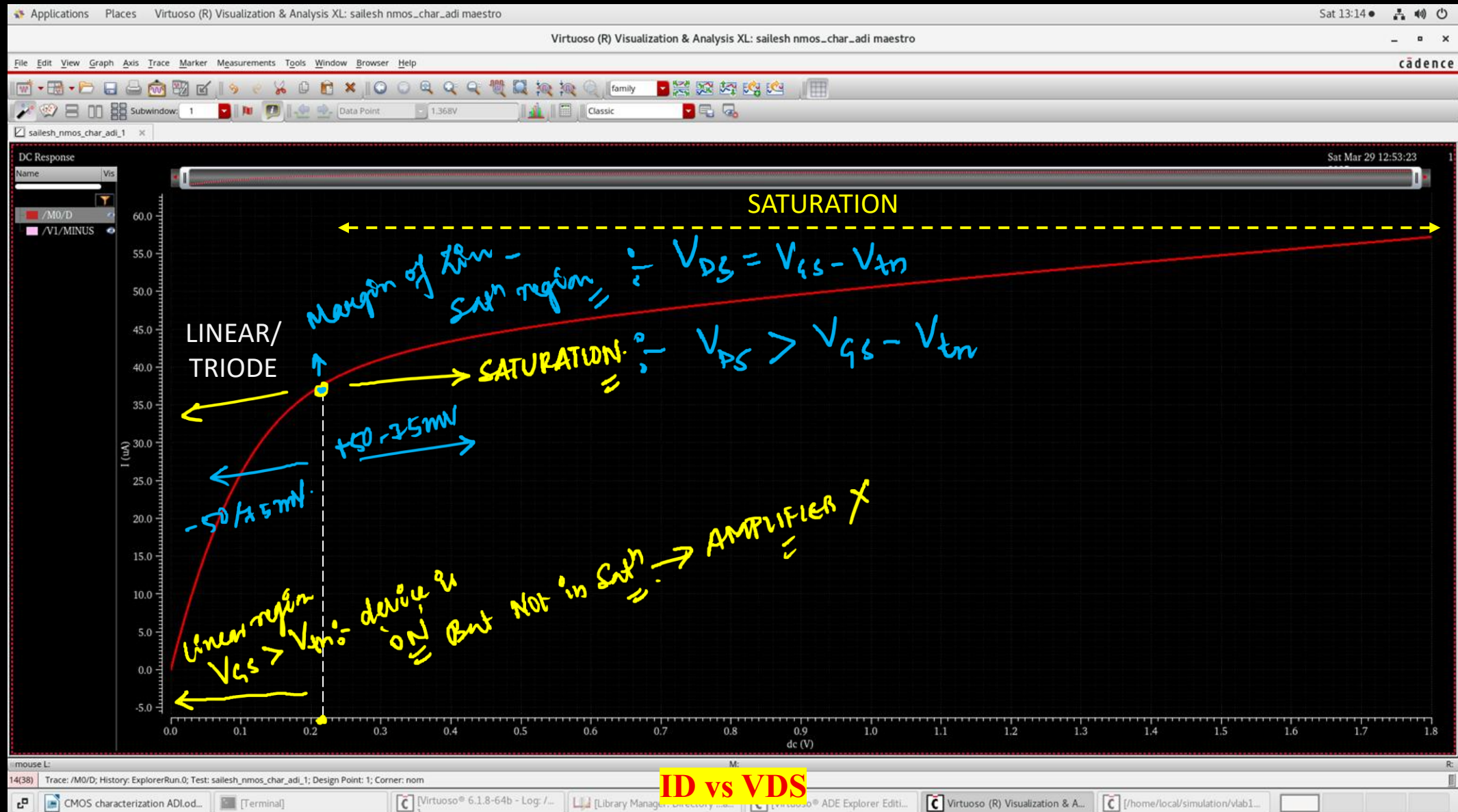


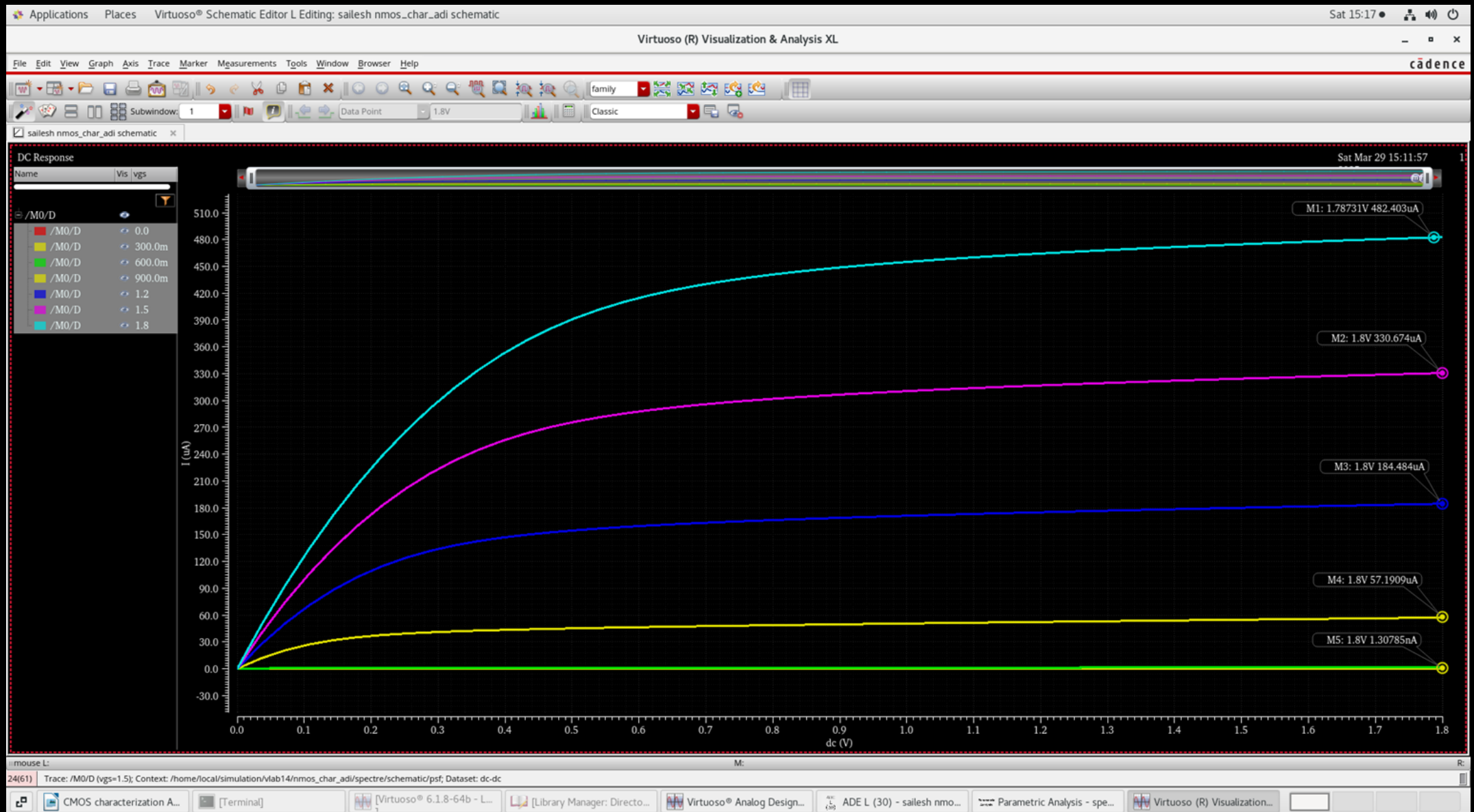
② $g_m = \frac{\Delta I_D}{\Delta V_{DS}}$
or g_{DS}

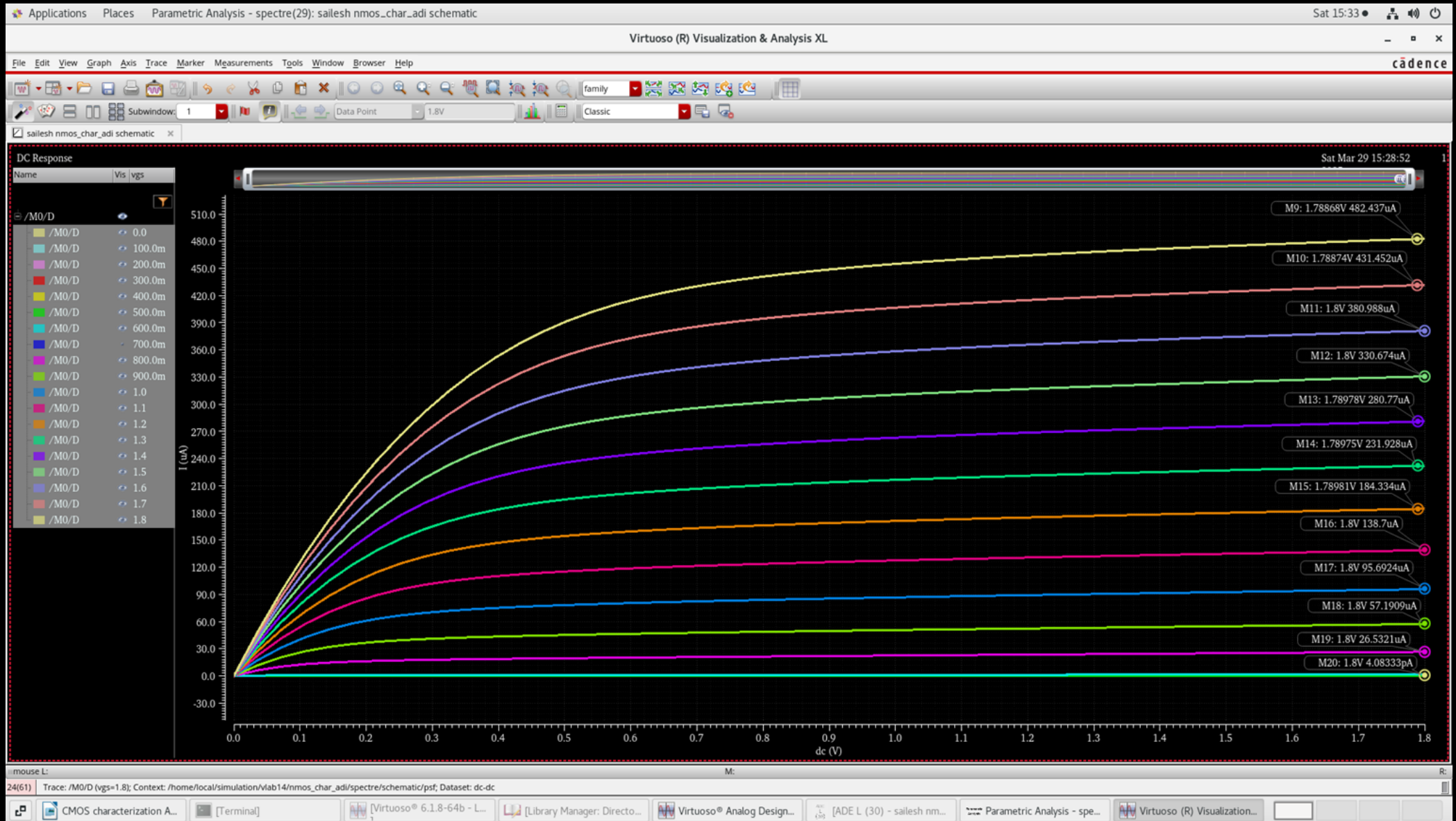


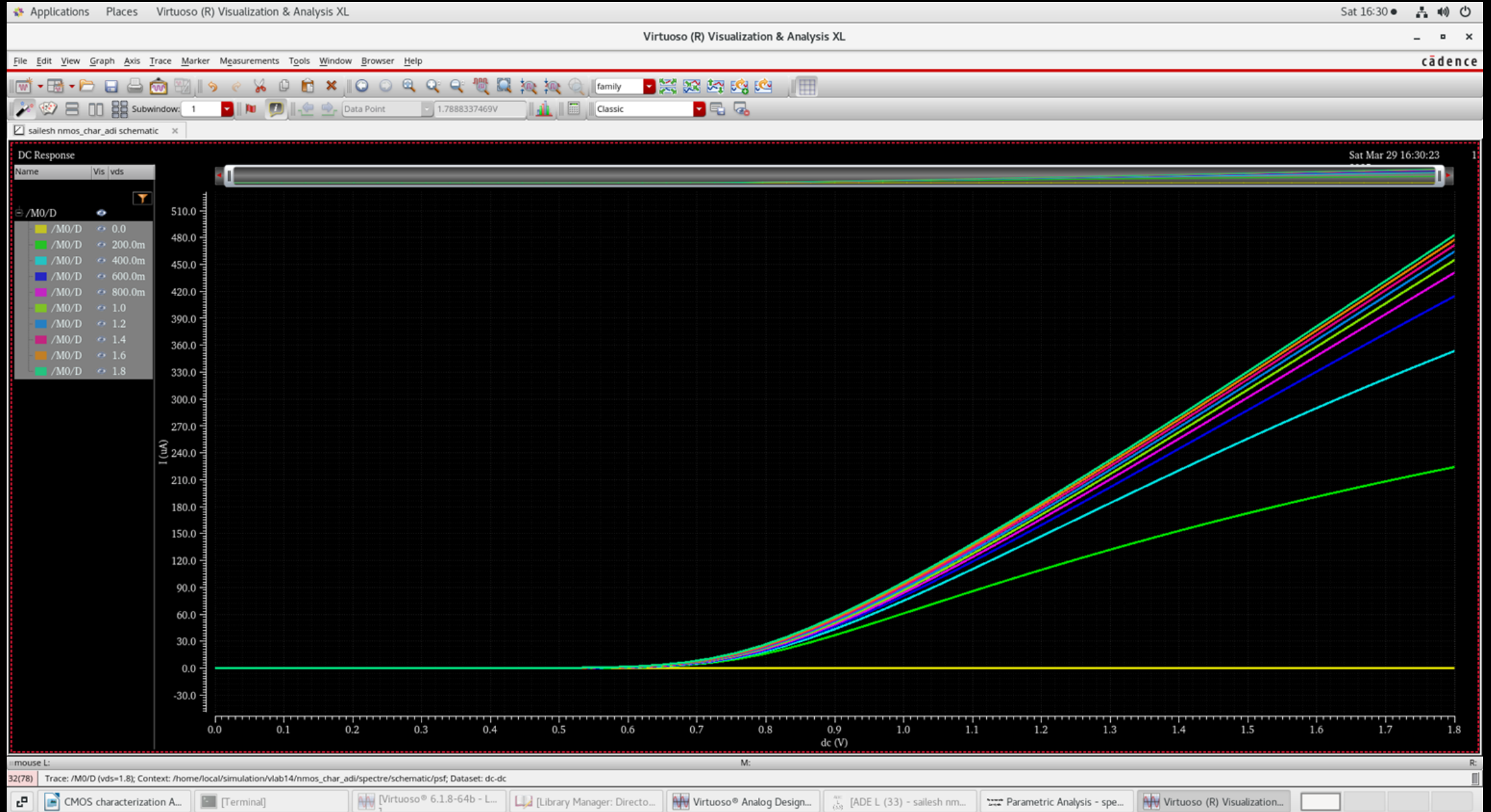
$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}} = \frac{10mA - 8mA}{V_{GS2} - V_{GS1}}$$

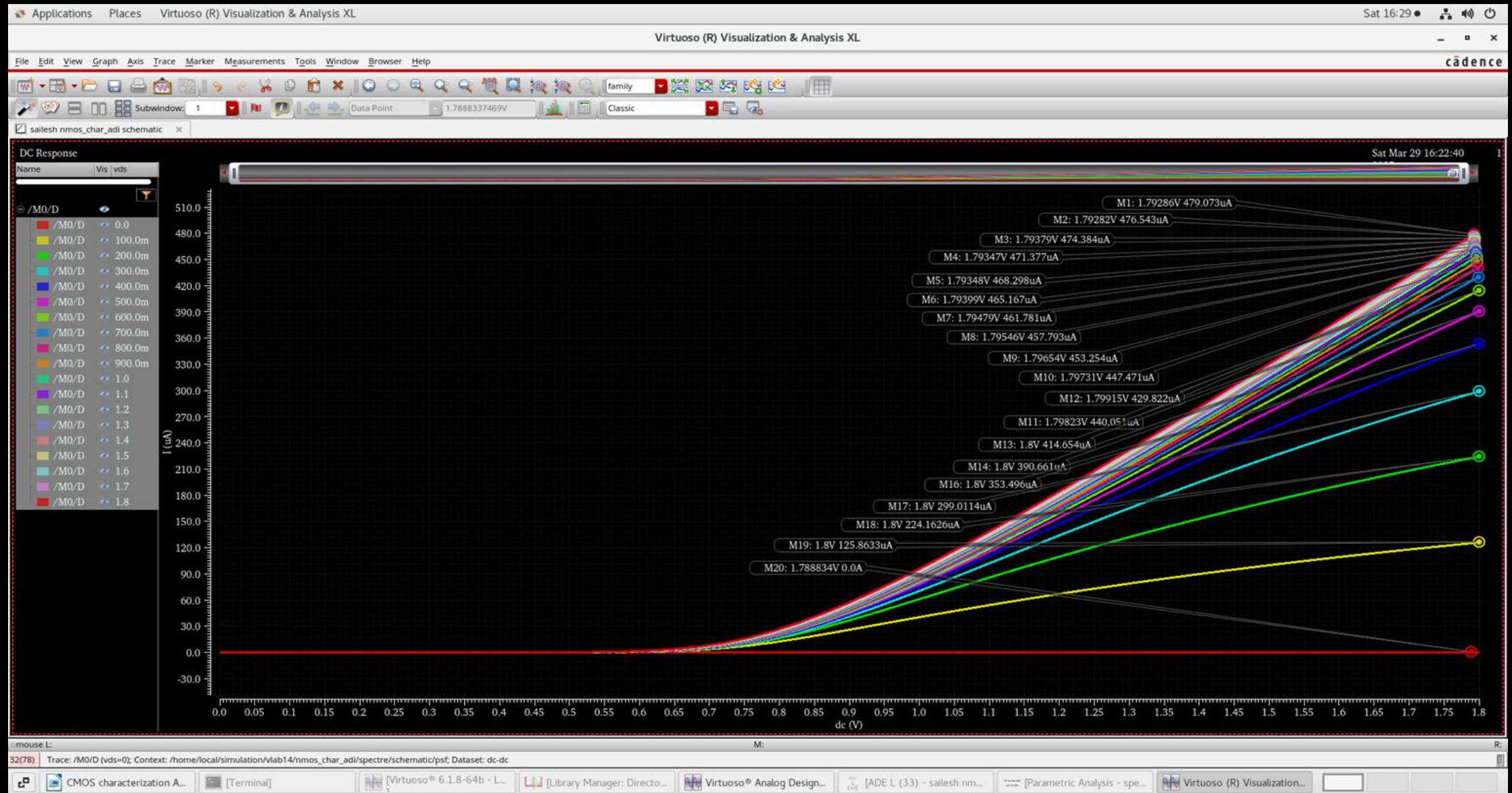




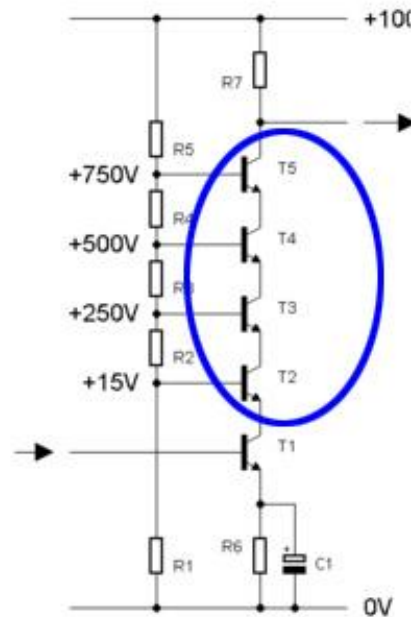




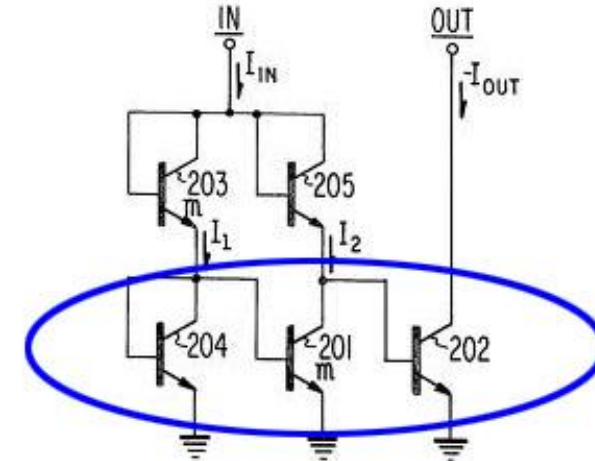




Cascoding Amp.
Transistor Loads are stacked
Vertically.



Cascading Amp.
Transistor Loads are
daisy chained horizontally.



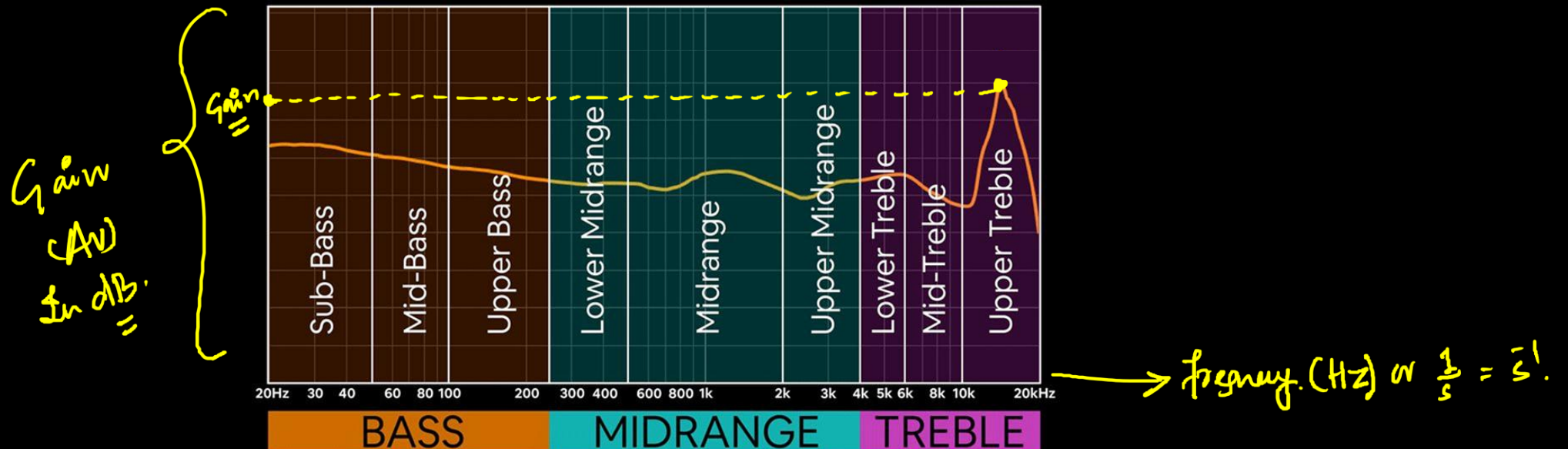
FREQUENCY RESPONSE

- The frequency response of a device is the characterization of its input-output relationship in the frequency domain, defined by the magnitude and phase of its transfer function $H(j\omega)$ which determines how signals of different frequencies are processed, amplified, or attenuated by the system.
- The frequency response of a system or device is the steady-state response of the system to a sinusoidal input signal as a function of frequency. Mathematically, it is obtained by evaluating the system's transfer function $H(s)$ along the imaginary axis, i.e.,

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

where:

- $X(j\omega)$ = Fourier Transform of the input signal
- $Y(j\omega)$ = Fourier Transform of the output signal
- $\omega = 2\pi f$ = angular frequency



■ Insights from the Graph . . .

1. Bass (20 Hz – 250 Hz)

- Sub-Bass (20–40 Hz): Deep rumble (like thunder, explosions). If weak → sound feels “thin.”
- Mid-Bass (40–80 Hz): Punch/kick (drums, bass guitar).
- Upper Bass (80–200 Hz): Warmth/body of sound.

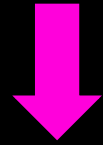
2. Midrange (250 Hz – 2 kHz)

- Lower Midrange (250–500 Hz): Fullness of vocals/instruments.
- Midrange (500 Hz – 1 kHz): Core detail of most instruments/voices.
- Upper Midrange (1–2 kHz): Clarity and presence of vocals.

3. Treble (2 kHz – 20 kHz)

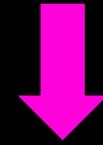
- Lower Treble (2–4 kHz): Definition/brightness. Too much → harsh. Too little → dull.
- Mid-Treble (4–8 kHz): Crispness, “airiness.”
- Upper Treble (8–20 kHz): Sparkle, fine details, cymbals, “shimmer.”

FREQUENCY RESPONSE



LOW FREQUENCY RESPONSE

- **Definition:** The low-frequency response of a device describes how the system behaves when the input signal frequency is near 0 Hz (DC) or in the lower end of the frequency spectrum. *Handwritten: $\omega \rightarrow 0$*
- Here, the output depends on how the device handles slowly varying signals or DC signals. *Handwritten: $f \downarrow$*
- **Mathematically:** It corresponds to the behavior of the transfer function $H(j\omega)$ as $\omega \rightarrow 0$. *Handwritten: $\omega \rightarrow 0 \rightarrow 2\pi f$; f is very less; ideally $f \rightarrow 0$; $\omega \rightarrow 0$.*
- **Example:**
 - In amplifiers, the low-frequency response is affected by coupling capacitors, bypass capacitors, leakage currents etc., which may cause attenuation at very low frequencies. *Handwritten: $I_s \rightarrow \mu A$*
 - In filters, the cut-off frequency determines where the low-frequency band starts being attenuated. *Handwritten: $f_c \propto \frac{1}{2\pi RC}$*
- Low-frequency roll-off occurs when the gain starts dropping at low frequencies.



HIGH FREQUENCY RESPONSE

- **Definition:** The high-frequency response of a device describes how the system behaves when the input signal frequency is very large (approaching MHz, GHz, etc., depending on the device). *Handwritten: $f \rightarrow \text{MHz, THz, ...}$*
- Here, the output depends on how the device handles **rapidly varying signals**.
- **Mathematically:** It corresponds to the behavior of the transfer function $H(j\omega)$ as $\omega \rightarrow \infty$. *Handwritten: $\omega \rightarrow \infty = 2\pi f$; $f \rightarrow \infty$*
- **Example:**
 - In amplifiers, the high-frequency response is limited by **parasitic capacitances**, **transistor transition frequency (f_T)**, **wiring inductances**.
 - In filters, beyond the **upper cut-off frequency**, the gain decreases, defining the **bandwidth limit**.
- **High-frequency roll-off** occurs when the gain starts decreasing due to these limitations.

LOW FREQUENCY RESPONSE

$$X_c = \frac{1}{2\pi f_c} = \frac{1}{f_c \rightarrow 0} = \frac{1}{0} = \infty \Rightarrow X_c = \infty \Omega$$

$$i = 0 A$$

$$0.0 C$$

- At low frequencies, the gain of a BJT amplifier is not constant because of the effect of coupling capacitors (C_s , C_c) and the emitter bypass capacitor (C_e).
- These capacitors form high-pass RC circuits, which block DC and attenuate very slow-varying signals. $\rightarrow f \rightarrow 0$
- As the frequency decreases, the capacitive reactance $X_C = 1/(2\pi fC)$ increases, reducing the effective signal transfer. from i/p to o/p.
- The lower cut-off frequency (f_L) is defined as the frequency where the gain falls to 70.7% (-3 dB) of its mid-band value.
- Below f_L , the amplifier shows low-frequency roll-off, reducing its ability to amplify signals accurately.

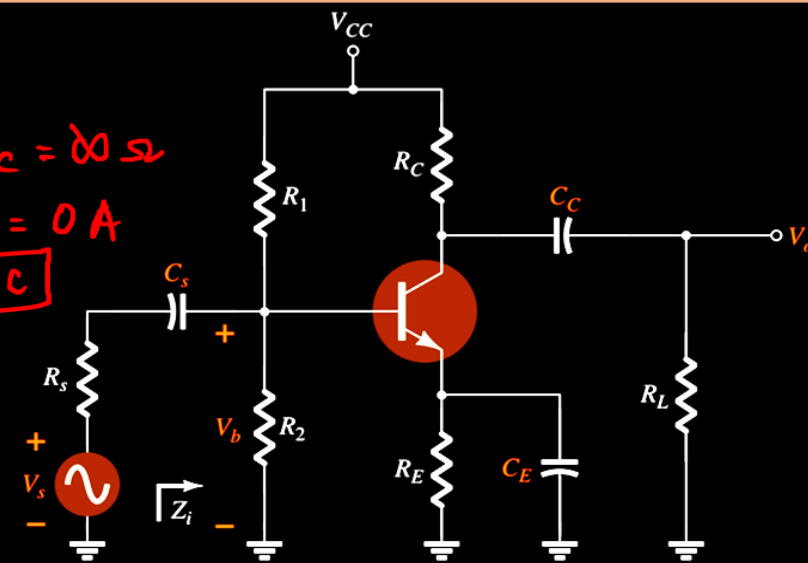


FIG. 9.32

Determining the effect of R_s on the low-frequency response of a BJT amplifier.

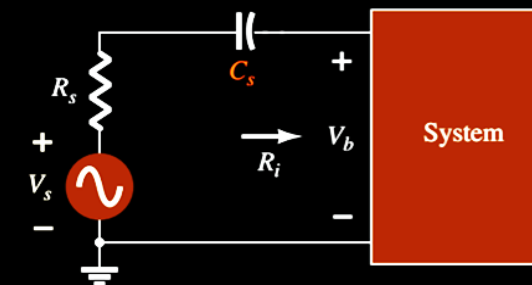


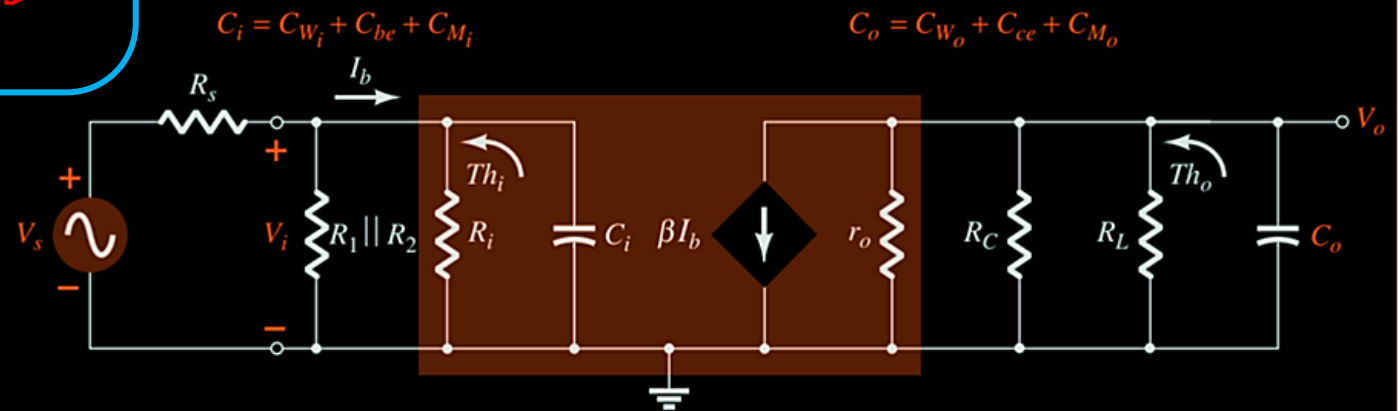
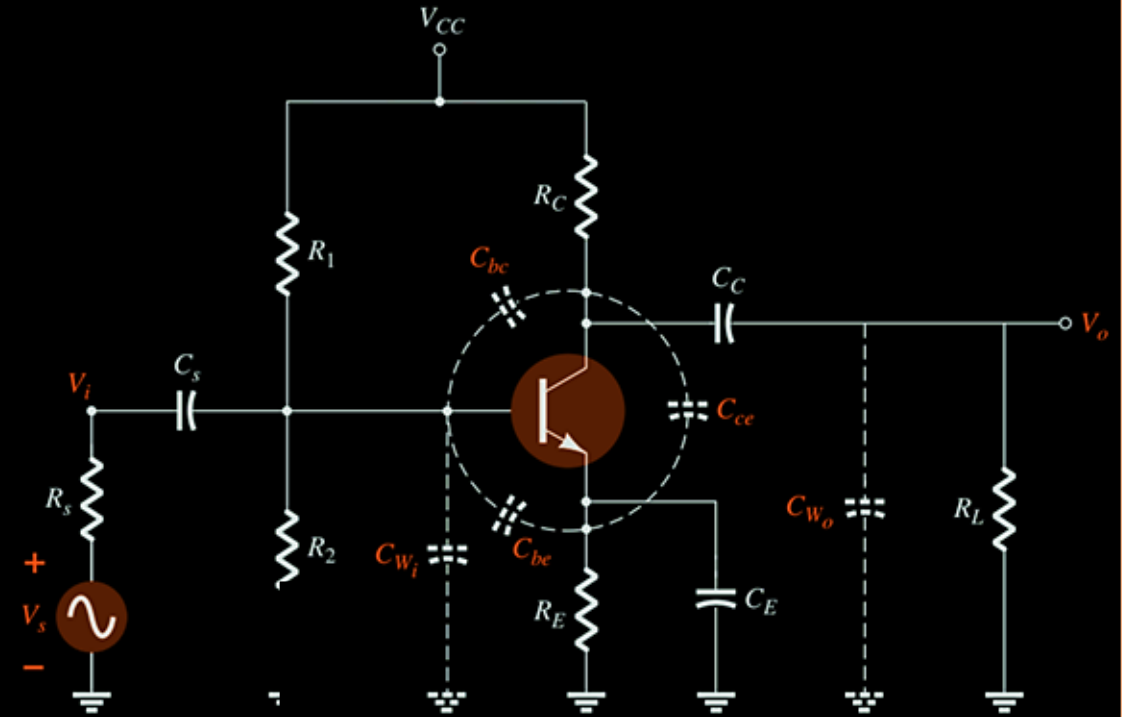
FIG. 9.33

Determining the effect of C_s on the low-frequency response.

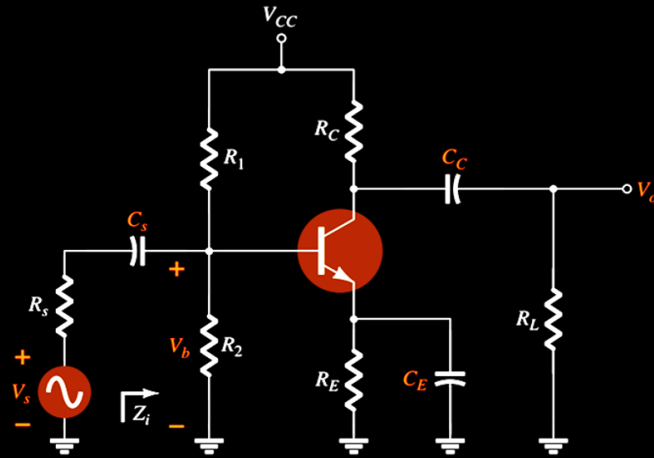
HIGH FREQUENCY RESPONSE

- At high frequencies, the amplifier is affected by internal transistor capacitances (C_{be} , C_{bc} , C_{ce}) and wiring capacitances.
- These parasitic capacitors create low-pass networks, attenuating high-frequency signals.
- The cut-off frequency (fH) depends on the input capacitance (C_i) and output capacitance (C_o).
- Above fH, the gain falls due to Miller effect and parasitic limitations, defining the upper roll-off.

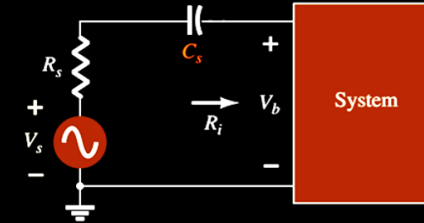
$$X_C = \frac{1}{2\pi f C} = \frac{1}{\infty} = 0 \Rightarrow X_C = 0 \left. \begin{array}{l} i = \infty \end{array} \right\} \boxed{\text{S.C.}}$$



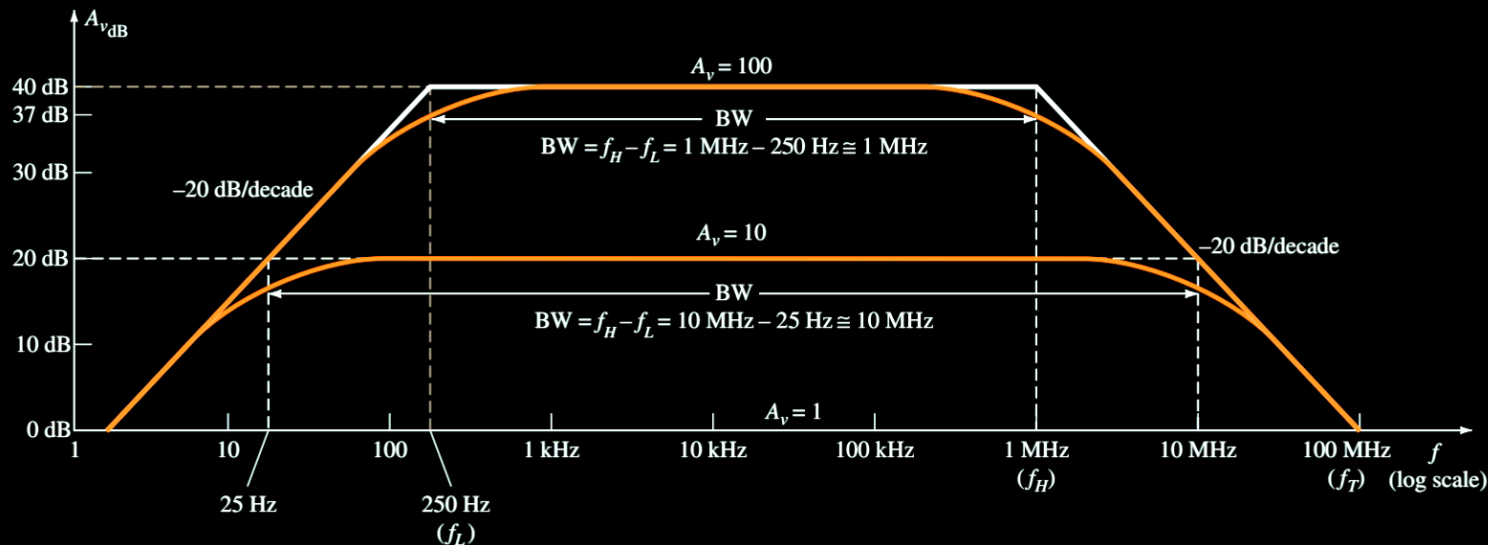
FREQUENCY RESPONSE OF BJT AMPLIFIER



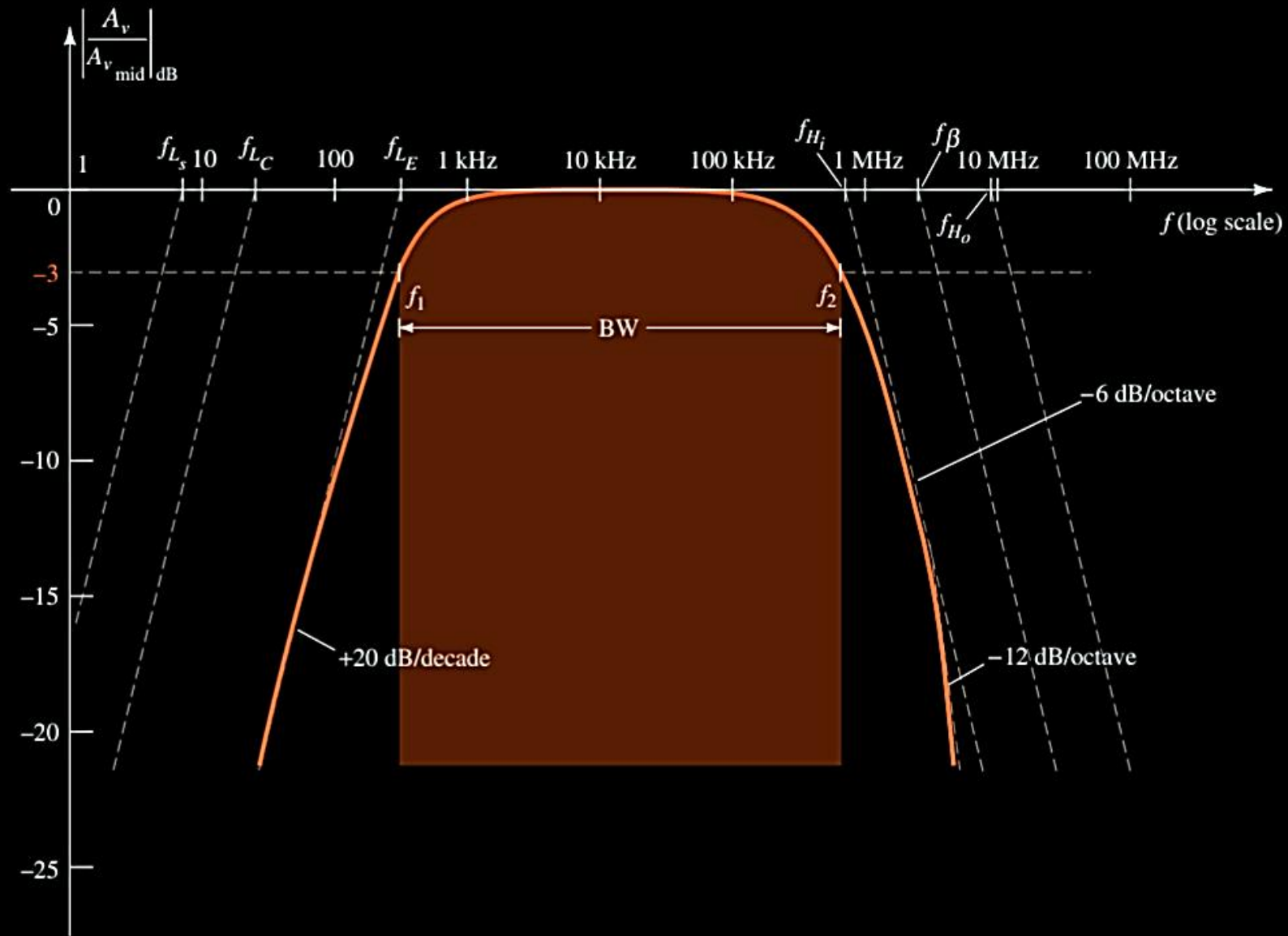
Determining the effect of R_s on the low-frequency response of a BJT amplifier.

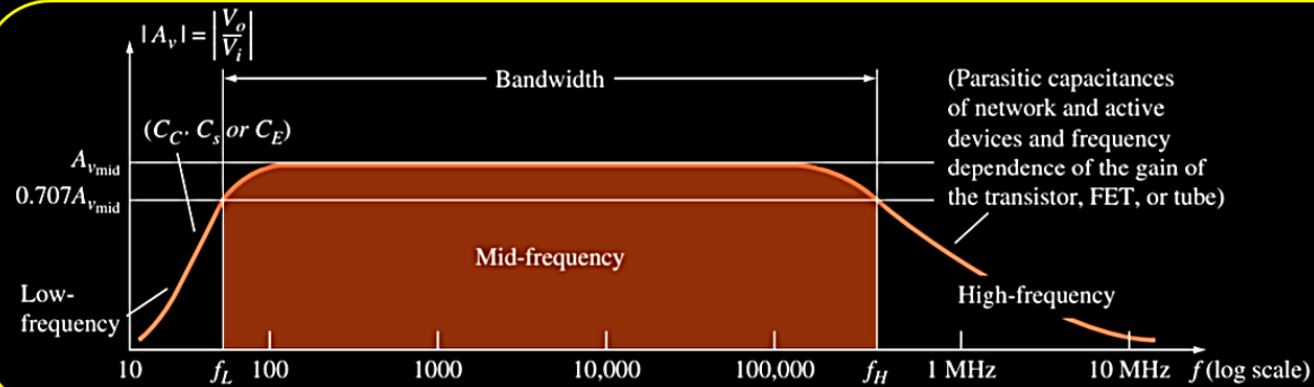


Determining the effect of C_s on the low-frequency response.

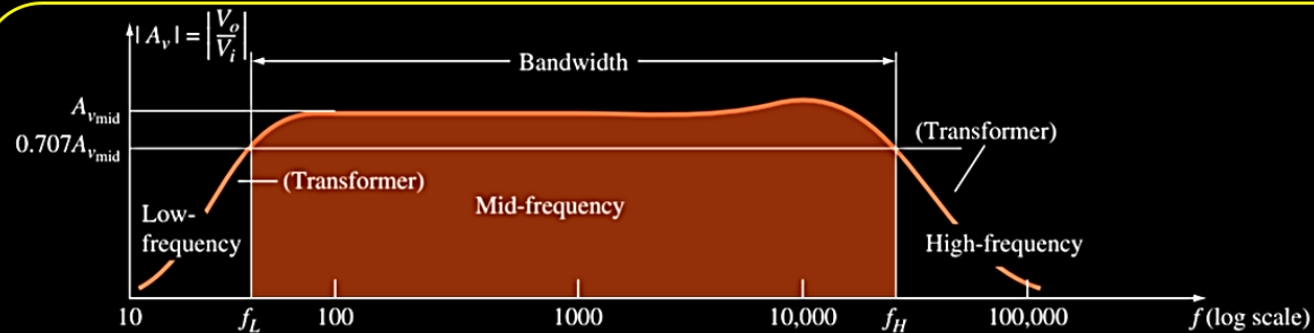


✳️ Complete R-C
Circuit simulⁿs.
= In LTSpice.

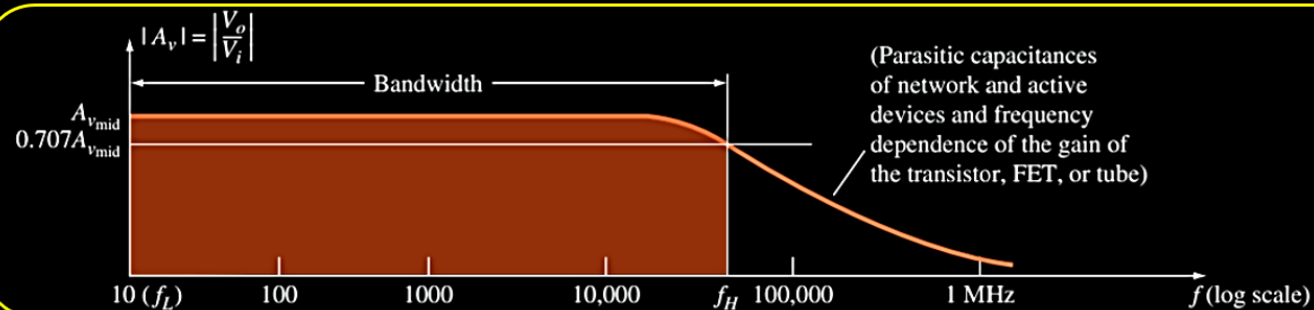




RC COUPLED AMPLIFIER

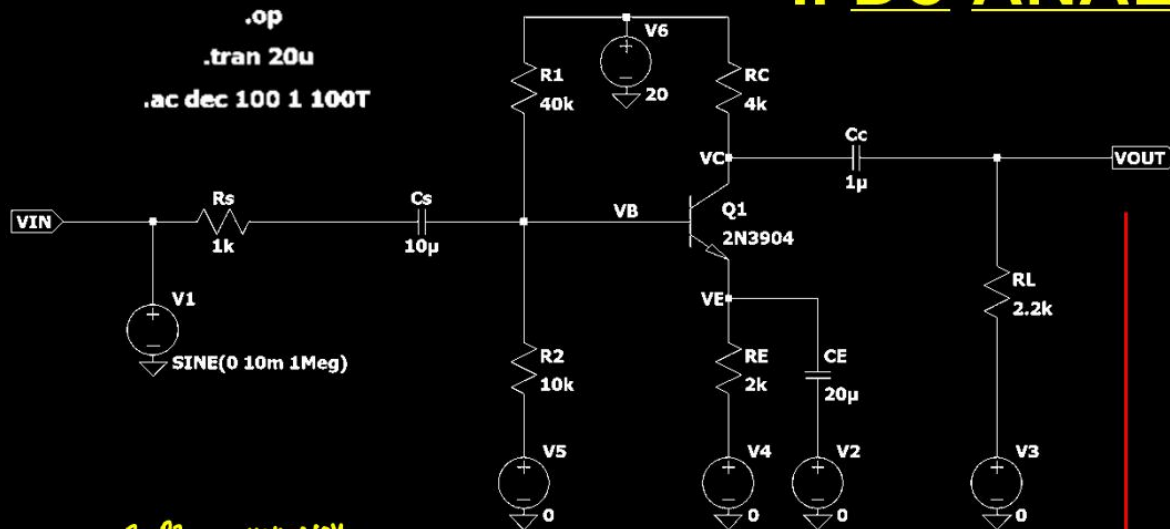


TRANSFORMER COUPLED AMPLIFIER



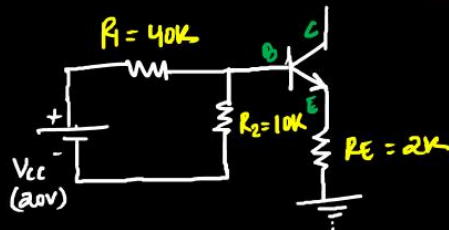
DIRECT COUPLED AMPLIFIER

I. DC ANALYSIS



$$R_{th} = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{40k \times 10k}{40k + 10k}$$

$$R_{th} = 8k \quad \checkmark$$

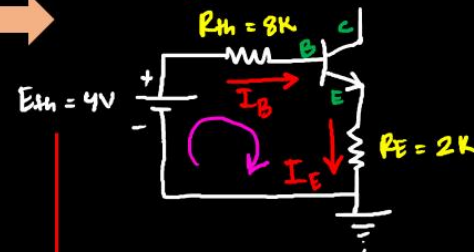


$$V_{th} = V_B = \frac{V_{cc} \cdot R_2}{R_1 + R_2} = \frac{20V \times 10k}{10k + 40k}$$

$$V_{th} = V_B = 4V > 0.7V$$

to TURN ON the BJT.

Putting KVL ; We get



$$V_{th} - I_B \cdot R_{th} - V_{BE} - I_E \cdot R_E = 0$$

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) \cdot R_E} = \frac{20V - 0.7V}{8k + (90 + 1) \cdot 2k} = \frac{19.3V}{8k + 182k} = 0.1015 \text{ mA}$$

$$\Rightarrow I_B = 5.78 \mu A \quad \checkmark$$

we know;

$$I_C = \beta \cdot I_B$$

$$= 90 \times 5.78 \mu A$$

$$= 520.2 \mu A$$

$$I_C = 0.5202 \text{ mA}$$

$$I_E = I_B + I_C$$

$$= 5.78 \mu A + 0.5202 \text{ mA}$$

$$= 0.52598 \text{ mA}$$

$$I_E = 0.525 \text{ mA}$$

$$\text{Again ; } V_{CE} = V_C - V_E = V_{cc} - I_C \cdot R_C - I_E \cdot R_E$$

$$= 20V - (0.5202 \text{ mA} \times 4k) - (0.525 \text{ mA} \times 2k)$$

$$V_{CE} = 20V - 2.08V - 1.05V$$

$$V_{CE} = 16.87V \quad \checkmark$$

--- Operating Point ---

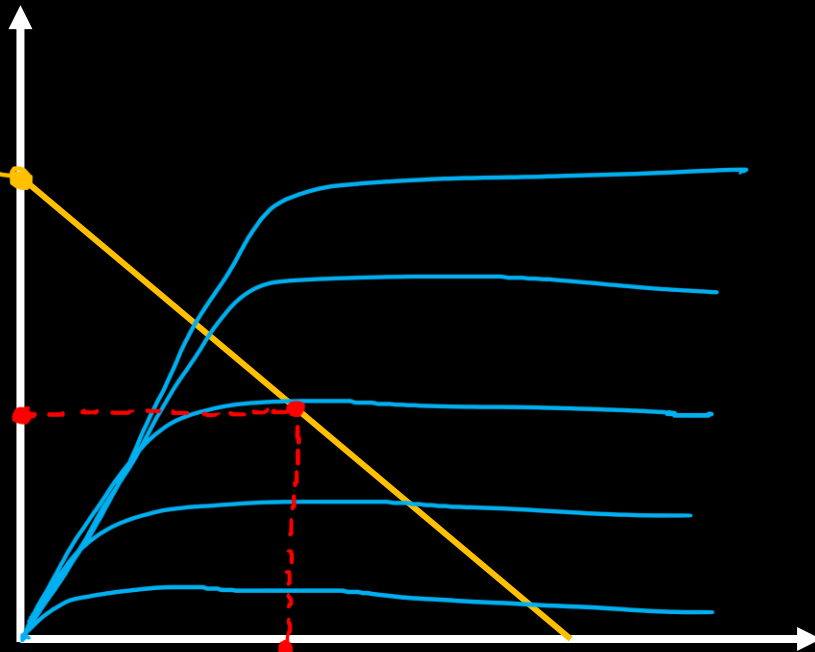
V(vb):	3.95985	voltage
V(n001):	20	voltage
V(n003):	0	voltage
V(n002):	3.95985e-14	voltage
V(n005):	0	voltage
V(vc):	13.4323	voltage
V(ve):	3.29389	voltage
V(n004):	0	voltage
V(vout):	2.95511e-14	voltage
V(n006):	0	voltage
V(vin):	0	voltage
Ic(Q1):	0.00164193	device_current
Ib(Q1):	5.01825e-06	device_current
Ie(Q1):	-0.00164694	device_current
Is(Q1):	0	device_current
I(RE):	0.00164694	device_current
I(RL):	1.34323e-17	device_current
I(V5):	0.000395985	device_current
I(V1):	3.95985e-17	device_current
I(R1):	0.000401004	device_current
I(Cs):	3.95985e-17	device_current
I(V3):	1.34323e-17	device_current
I(V6):	-0.00204293	device_current
I(R2):	0.000395985	device_current
I(Cc):	-1.34323e-17	device_current
I(RC):	0.00164193	device_current
I(CE):	6.58778e-17	device_current
I(V2):	6.58778e-17	device_current
I(V4):	0.00164694	device_current
I(Rs):	3.95985e-17	device_current

$$\frac{V_{CC}}{R_C + R_E} = \frac{20V}{4k + 2k} = 3.33 \text{ mA}$$

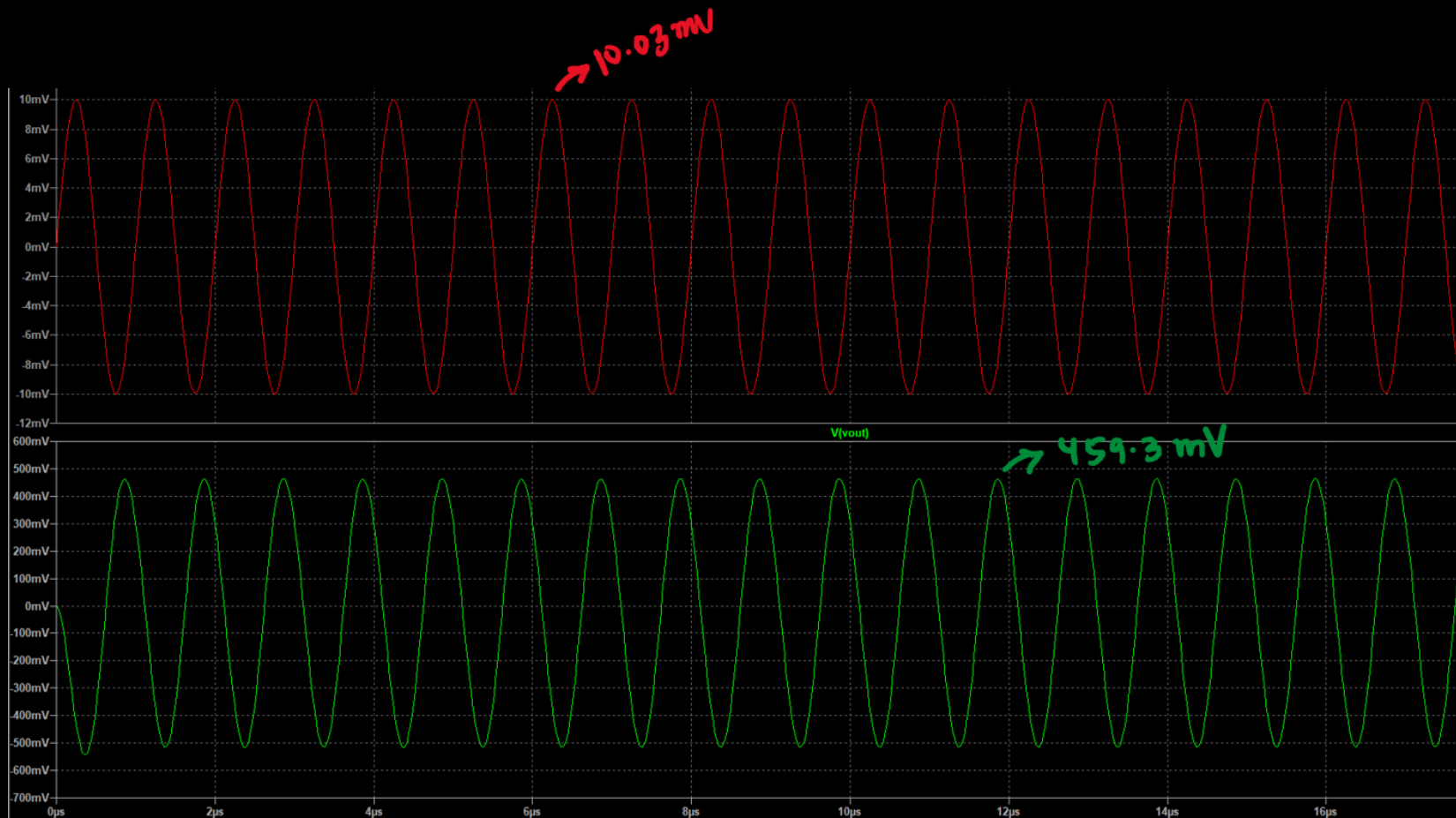
$$(I_C)_Q = 0.5202 \text{ mA}$$

$$(V_{CE})_Q = 16.87V$$

V_{CC}
(20V)



II. TRANSIENT ANALYSIS



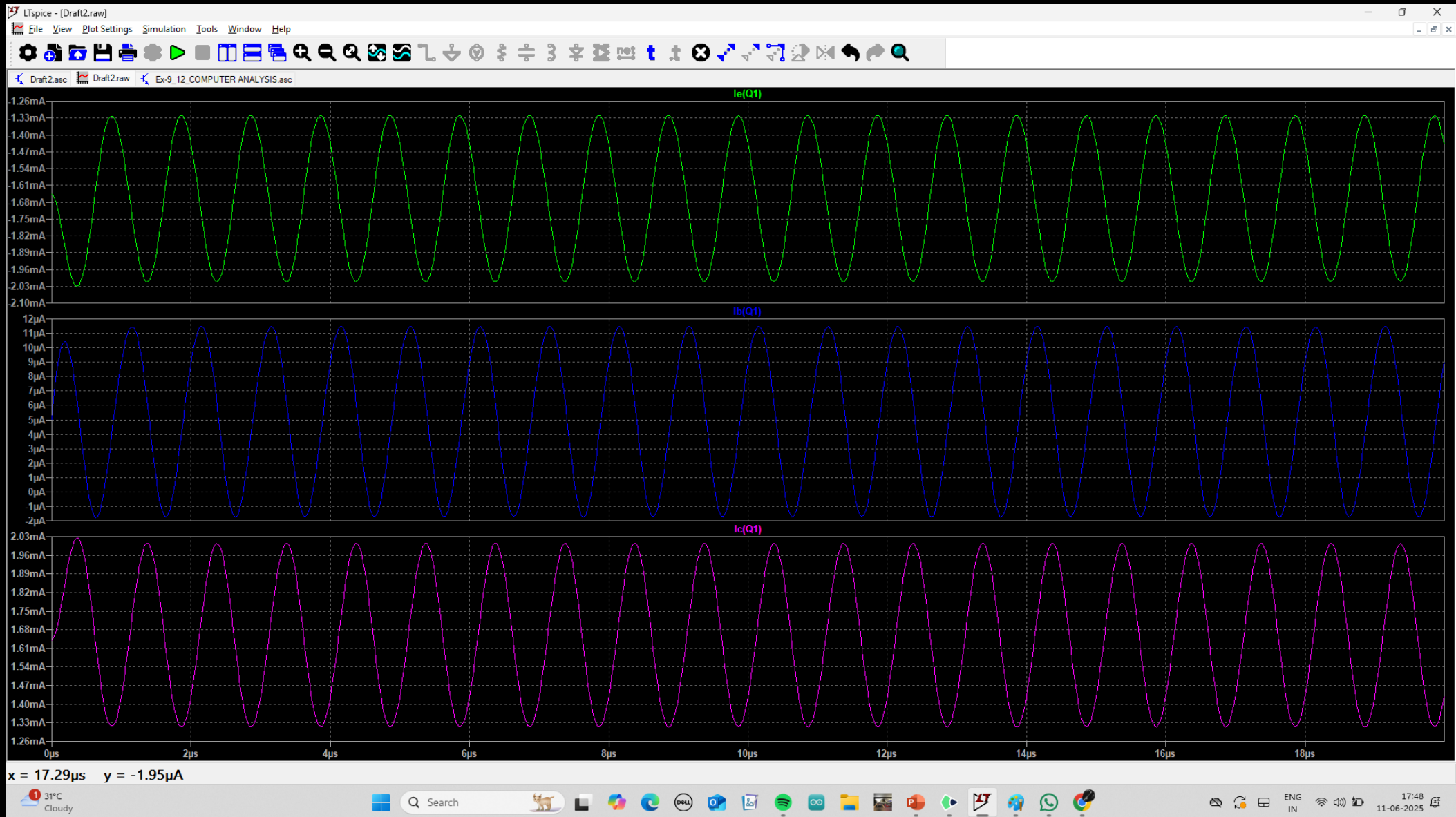
$$A_v = \text{voltage gain} = \frac{(v_{out})_{pp}}{(v_{in})_{pp}} =$$

$$A_v = \frac{462.25 \text{ mV} + 514.53 \text{ mV}}{9.992 \text{ mV} + 9.994 \text{ mV}}$$

$$A_v = 48.8732$$

$$A_v = \frac{v_{out}}{v_{in}}$$

$$v_{out} = A_v \times v_{in}^{**}$$



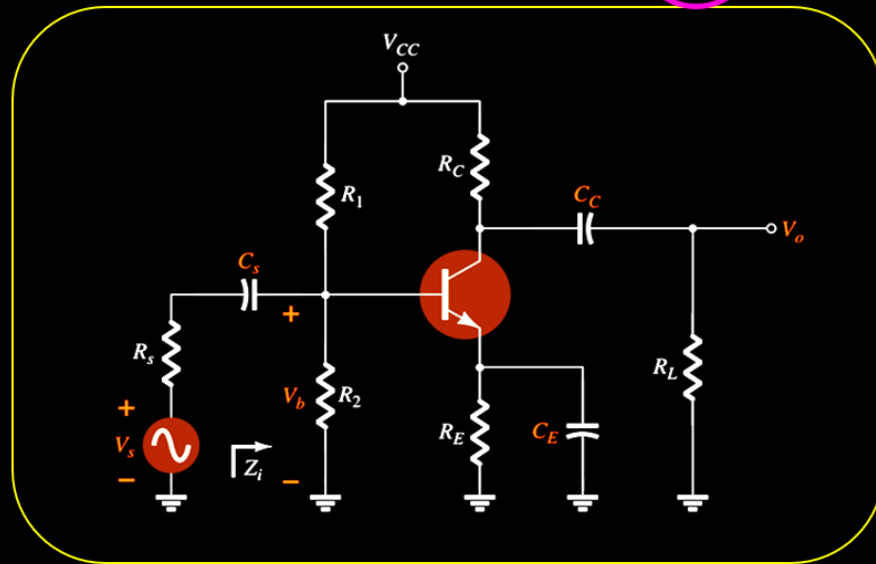
III. AC ANALYSIS



LOW FREQUENCY
RESPONSE

For
Finding the
LOW CUT-OFF FREQUENCY

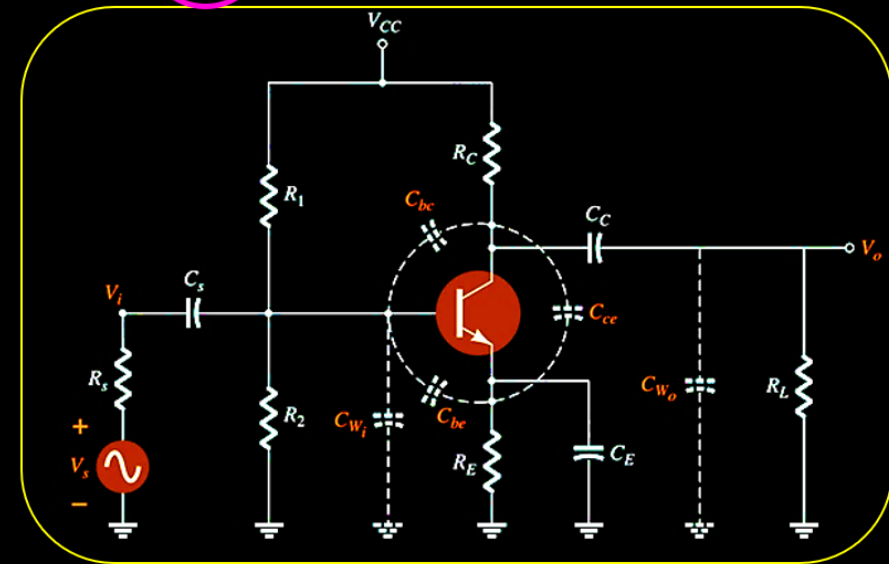
f_L

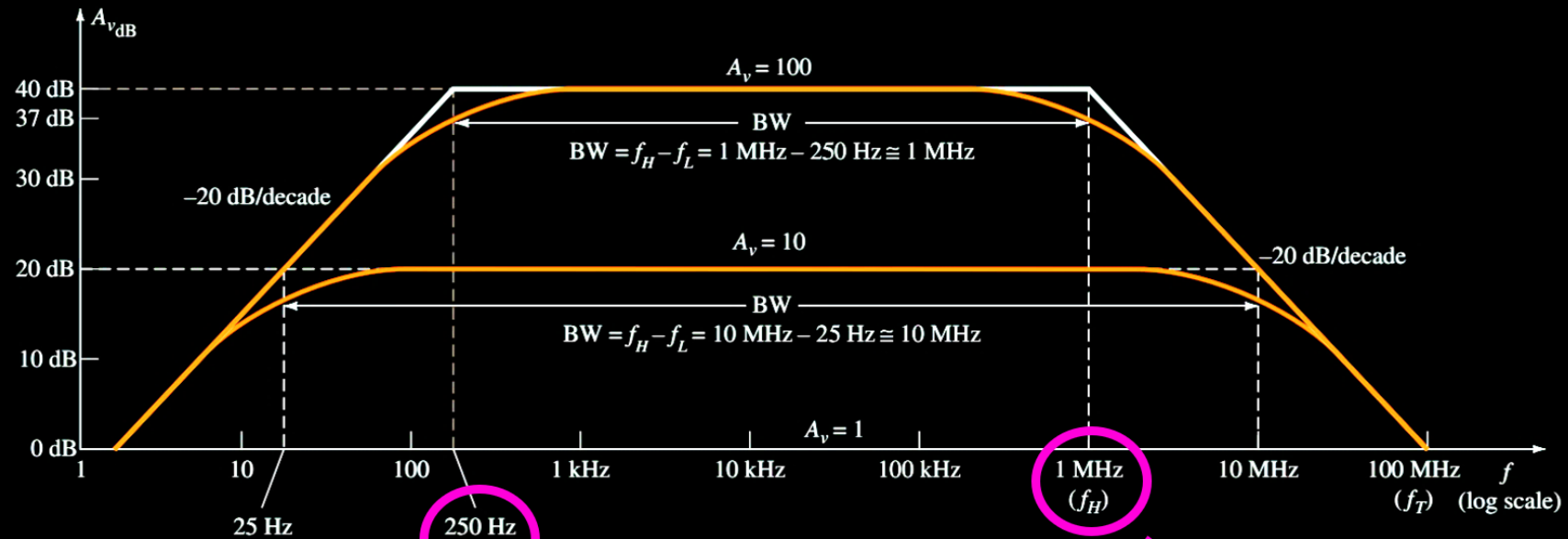


HIGH FREQUENCY
RESPONSE

For
Finding the
HIGH CUT-OFF FREQUENCY

f_H





1. Due to Source Capacitor

$$f_{L_s} = \frac{1}{2\pi(R_s + R_i)C_s}$$

2. Due to By-pass Emitter Capacitor

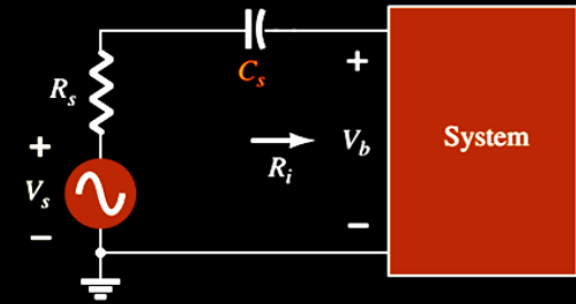
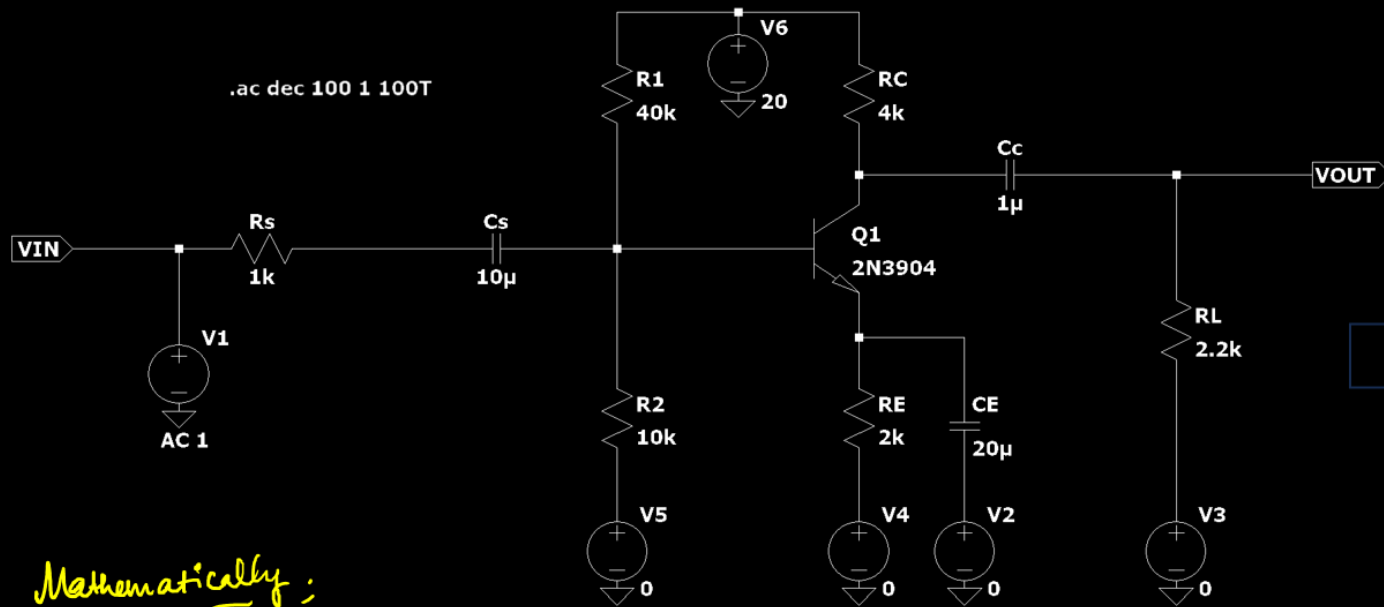
$$f_{L_E} = \frac{1}{2\pi R_e C_E}$$

3. Due to Output Coupling Capacitor

$$f_{L_C} = \frac{1}{2\pi(R_o + R_L)C_C}$$

$$f_{H_i} = \frac{1}{2\pi R_{Th_i} C_i}$$

$$f_{H_o} = \frac{1}{2\pi R_{Th_o} C_o}$$



Mathematically ;

$$A_v = \text{voltage gain} = \frac{V_o}{V_i} = -\frac{R_c \parallel R_L}{r_e} = \frac{R_c \times R_L}{R_c + R_L} \times \frac{1}{r_e} = R_c \parallel R_L \times \frac{1}{V_T / I_E} = R_c \parallel R_L \times \frac{I_E}{V_T} = \frac{1}{V_T} [R_c \parallel R_L \times I_E]$$

$$A_v \propto R_c \cdot R_L \propto I_E \propto I_S$$

$$= I_B + I_C$$

$$= I_B + I_S \cdot e^{\frac{V_{BE}}{V_T}}$$

Since; $V_T = 0.0259 \text{ V} \approx 26 \text{ mV}$ is constant ✓

$$I_C \propto I_S \propto e^{V_{BE}/V_T}$$

$$e^1 = 2.7182 ; e^2 = 7.389$$

That Means; if $e^{V_{BE}} \uparrow \rightarrow I_S \uparrow \rightarrow I_C \uparrow \rightarrow I_E \uparrow \rightarrow r_e \downarrow \rightarrow A_v \uparrow \dots$

So; According to the Model parameters of NXP Semi-conductor's $\rightarrow$ BJT Model Name: "2N3904"

$$\beta = 90$$

$$I_C = ?$$

$$I_C = I_S \cdot e^{V_{BE}/V_T}$$

$$V_{BE} = V_T \cdot \ln\left(\frac{I_C}{I_S}\right) = 0.0259V \cdot \ln\left(\frac{0.5202mA}{10fA}\right) = \boxed{0.692V}$$

we calculated taking $V_{BE} = 0.7V$.

$$\boxed{V_{BE} = 0.692V \text{ or } 692mV}$$

But According to our DC Biased operating point:-

$$V_{th} = V_{CC} \cdot \frac{R_2}{R_1 + R_2} = \frac{20V \cdot 10K}{40K + 10K} = \frac{200}{50} = \boxed{4V}$$

$$\boxed{V_{th} = V_B = 4V}$$

$[V_{th} \text{ or } V_B > V_{BE}] \rightarrow$ Hence; Transistor is 'ON' !!

$$V_{CE} = 16.87V$$

$$I_C = 0.5202mA$$

SUFFICIENTLY HIGH ✓
Hence [ACTIVE REGION]

Now; Let's put the values and find out $\bar{A}_v$ [voltage gain].

$$A_v = \frac{V_o}{V_i} = -\frac{R_C \parallel R_L}{r_e} = \frac{1}{V_T} \cdot [(R_C \parallel R_L) \cdot I_E]$$

$$\begin{aligned} & \frac{1}{0.0259V} \cdot \left[\frac{(R_C \times R_L)}{R_C + R_L} \cdot (I_E) \right] \\ &= \frac{1}{0.0259V} \cdot \left[\frac{4K \times 2.2K}{4K + 2.2K} \right] \cdot (1.654mA) \\ &= \end{aligned}$$